6. Construct a C Program to Simulate Round Robin Scheduling Algorithm with C

Code - Part 1 (Lines 1–44)

```c
1  #include <stdio.h>
2
3  int main()
4  {
5      int i, NOP, sum = 0, count = 0, y, quant;
6      int wt = 0, tat = 0;
7      int at[10], bt[10], temp[10];
8      float avg_wt, avg_tat;
9
10     printf("Total number of process in the system: ");
11     scanf("%d", &NOP);
12
13     y = NOP;
14
15     for(i = 0; i < NOP; i++)
16     {
17         printf("\nEnter Arrival Time and Burst Time for Process[%d]\n", i + 1);
18
19         printf("Arrival Time: ");
20         scanf("%d", &at[i]);
21
22         printf("Burst Time: ");
23         scanf("%d", &bt[i]);
24
25         temp[i] = bt[i];
26     }
27
28     printf("\nEnter Time Quantum: ");
29     scanf("%d", &quant);
30
31     printf("\nProcess No\tBurst Time\tTAT\tWaiting Time");
32
33     for(sum = 0, i = 0; y != 0;)
34     {
35         if(temp[i] <= quant && temp[i] > 0)
36         {
37             sum = sum + temp[i];
38             temp[i] = 0;
39             count = 1;
40         }
41         else if(temp[i] > 0)
42         {
43             temp[i] = temp[i] - quant;
44             sum = sum + quant;
```

Code - Part 2 (Lines 45–78)

```
45          }
46
47          if(temp[i] == 0 && count == 1)
48          {
49              y--;
50
51              printf("\nP[%d]\t\t%d\t\t%d\t%d",
52                      i + 1,
53                      bt[i],
54                      sum - at[i],
55                      sum - at[i] - bt[i]);
56
57              wt += (sum - at[i] - bt[i]);
58              tat += (sum - at[i]);
59
60              count = 0;
61          }
62
63          if(i == NOP - 1)
64              i = 0;
65          else if(at[i + 1] <= sum)
66              i++;
67          else
68              i = 0;
69      }
70
71      avg_wt = (float)wt / NOP;
72      avg_tat = (float)tat / NOP;
73
74      printf("\n\nAverage Turnaround Time: %.2f", avg_tat);
75      printf("\nAverage Waiting Time: %.2f\n", avg_wt);
76
77      return 0;
78  }
```

Output

```
Total number of process in the system: 1

Enter Arrival Time and Burst Time for Process[1]
Arrival Time: 0
Burst Time: 5

Enter Time Quantum: 2

Process No      Burst Time      TAT     Waiting Time
P[1]            5               5       0

Average Turnaround Time: 5.00
Average Waiting Time: 0.00


...Program finished with exit code 0
Press ENTER to exit console.
```